

Assignment 3 Response Sheet

2026 Semester 1

1 Instructions

Submission deadline: 26 April 2026 at 23:00. (Week 8, Sunday) Please type your answers in the spaces provided. Do not repeat the questions in your answers. All tasks are compulsory.

Note that you must correctly cite and reference any sources you have consulted. You may use any internationally recognized referencing style such as APA or IEEE referencing.

Submit this document to the “Assignment 3 Main Submission” inbox by the published deadline.

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2 Task 1: Literature Review

This literature review examines whether in-context demonstration design can support robust and faithful out-of-distribution generalisation for tabular prediction tasks. The review proceeds in five stages. It first formalises the types of distribution shift and the benchmarks that operationalise them, and then traces why models fail under shift through the shortcut-learning literature. It subsequently situates the project within the in-context learning and tabular LLM literatures, reviews the evolution from invariance-based training objectives to faithfulness-based evaluation, and concludes by examining the demonstration-selection literature and the gap it leaves open. The proposed project evaluates ICL under tabular distribution shift, and the careful distinction between covariate, label, and concept shift is necessary for interpreting whether a given demonstration strategy is genuinely robust or merely effective under one benign shift type.

2.1 OOD generalization and its forms

The study of generalization under distribution shift predates the modern machine learning literature, having long been a concern in statistics and econometrics under names such as covariate shift, sample selection bias, and concept drift. Each community developed its own terminology, which has historically rendered cross-comparison difficult [1]. Despite this fragmentation, the underlying mathematical object is consistent across formulations: a divergence between the joint distribution from which training data was drawn and the joint distribution under which the model is deployed.

2.1.1 What OOD means formally

Out-of-distribution (OOD) generalization refers to the ability of a model trained on a source distribution to perform well on a target distribution that differs from it. Following the notation in [2], an OOD setting is one in which $P_{tr}(x, y) \neq P_{te}(x, y)$, where P_{tr} and P_{te} denote the training and test joint distributions over inputs x and targets y .

The taxonomy of dataset shift is constructed by asking which factor of the joint distribution moves. [1] consolidate the literature around three principal categories, defined here for the $X \rightarrow Y$ (causal) direction:

- **Covariate shift:** the marginal distribution of inputs changes while the conditional distribution of the target given the input remains stable. Formally, $P_{tr}(x) \neq P_{te}(x)$ but $P_{tr}(y | x) = P_{te}(y | x)$.
- **Prior probability shift:** the class prior changes while the class-conditional input distribution remains stable. Formally, $P_{tr}(y) \neq P_{te}(y)$ but $P_{tr}(x | y) = P_{te}(x | y)$. This category is most naturally defined for $Y \rightarrow X$ problems.
- **Concept shift:** the marginal input distribution remains constant but the functional relationship between inputs and targets changes. Formally, $P_{tr}(x) = P_{te}(x)$ but $P_{tr}(y | x) \neq P_{te}(y | x)$.

These three cases do not exhaust the possibilities; sample selection bias, domain shift, and source component shift constitute further refinements catalogued by [3]. Nevertheless, the three categories enumerated above form the canonical vocabulary employed in subsequent benchmark and method papers.

2.1.2 Why these distinctions matter for modelling choices

The taxonomy is not descriptive pedantry: each shift type corresponds to a distinct causal structure, and the appropriate correction strategy depends on which structure is in play. [3] develops this argument by pairing each shift type with a probabilistic graphical model and the modelling response it admits. A central observation is that purely conditional models satisfying Kolmogorov consistency, of which Gaussian processes are the canonical example, are in principle

invariant to covariate shift, since their predictive distribution does not depend on the marginal $P(x)$. Models such as the support vector machine, by contrast, do not satisfy this consistency property and are therefore sensitive to covariate density even when the conditional $P(y | x)$ is unchanged. This distinction demonstrates that whether covariate shift requires explicit correction is itself a model-class question rather than a property of the shift alone.

The methodological consequence is that methods designed for one shift type can underperform, or even actively harm performance, under another. For prior probability shift, the appropriate response is to replace the training prior with the (estimated) test prior in a generative model, while leaving the class-conditional input model untouched [3]. For concept shift, where the conditional $P(y | x)$ itself moves, no reweighting of training data can recover the test relationship without additional information about the shift mechanism. Misidentifying the shift type therefore risks deploying a correction that is at best inert and at worst counterproductive.

Recent empirical work has sharpened this point. [4] re-evaluate a wide range of domain-generalization algorithms under consistent experimental conditions in the DomainBed testbed and find that, when properly tuned, empirical risk minimization is competitive with or outperforms more sophisticated methods, with no algorithm exceeding ERM by more than one accuracy point on average. Their interpretation is that better in-distribution generalization is a dominant factor behind better OOD generalization, and that previously reported gains from specialized methods may have been artefacts of inconsistent tuning. The implication for the present review is twofold: first, empirical claims about OOD methods require disciplined model selection in order to be credible; and second, the choice of method must be justified against the specific shift structure being addressed, rather than against an undifferentiated notion of “distribution shift.”

2.1.3 Benchmarks that operationalise OOD

Translating the taxonomy into reproducible empirical study requires benchmarks that fix specific source and target distributions and specify how shift is measured. The image and text modalities possess a mature ecosystem of such benchmarks, whereas the tabular setting has historically lagged behind [5]. Recent work has begun to close this gap by introducing benchmarks that map the shift types defined above onto concrete evaluation protocols.

The most prominent of these is **TableShift** [6], a benchmarking suite purpose-built for distribution shift in tabular data. It comprises 15 binary classification tasks drawn from domains including finance, education, public policy, healthcare, and civic participation. Each task is paired with a meaningful and naturally occurring shift, such as predicting hospital readmission across patient demographics, ICU mortality across hospitals, or income across geographic regions. Beyond the tasks themselves, **TableShift** contributes a metric suite that maps directly back to the shift taxonomy of [2]:

- **In-Distribution (ID) and Out-of-Distribution (OOD) Test Accuracy:** the headline performance numbers reported on the source and shifted target domains respectively.
- **Shift Gap (ΔAcc):** the difference between ID and OOD accuracy, which directly quantifies the cost imposed by the shift.
- **Optimal Transport Dataset Distance (OTDD):** employed to measure covariate shift (Δx) by quantifying movement in the input feature distribution across domains.
- **Frechet Dataset Distance (FDD):** which uses the Wasserstein-2 distance on classifier feature activations to approximate concept shift ($\Delta y | x$).
- **Base Rate L_2 Difference:** employed for label shift (Δy), computed between the empirical class means of the source and target distributions.

The value of these metrics is that they enable researchers not only to measure how badly a model degrades, but also to attribute that degradation to a specific kind of shift, which in turn informs the modelling and correction strategies discussed in the previous subsection.

Complementing this, **WHYSHIFT** [7] provides 22 settings constructed from open-source tabular datasets such as Taxi, US Accident, and the ACS datasets. Its particular focus is on concept shift, which is generally regarded as the **most difficult regime** to correct for because the input-output mapping itself changes between train and test. By isolating distinct shift patterns within a **single benchmark**, WHYSHIFT enables practitioners to test whether methods that perform well under covariate shift continue to hold up once the underlying conditional $P(y | x)$ also moves.

A more recent addition, motivated by the rise of large language models (LLMs) as general-purpose tabular reasoners, is **STaDS** (Structured Tabular Decision Simulations) [8]. STaDS spans 15 real-world tabular datasets across healthcare, finance, and public policy, and explicitly argues that accuracy is an **insufficient signal of model competence** in this setting because a model can produce the **correct label while reasoning over the wrong features**. The full evaluation protocol it proposes, comprising predictive competence, comprehension fidelity, and decision faithfulness, is discussed in detail in Section 2.5, where it **forms the foundation of the faithfulness-as-evaluation strand of the literature**.

Together, these three benchmarks cover the spectrum of shift types this project addresses: **TableShift** provides a broad operationalisation of covariate, label, and concept shift on natural tabular tasks; **WHYSHIFT** isolates the harder concept-shift regime; and **STaDS** supplies the evaluation infrastructure required when accuracy alone is not a trustworthy proxy for understanding. The remaining gap, which motivates the rest of this review, is that **none of these benchmarks directly examines whether *in-context* adaptation in LLMs can produce robust**

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generalization on tabular tasks, and whether such generalization is achieved for the right reasons.

2.2 Why OOD fails: shortcut learning and spurious correlations

2.2.1 Models prefer simple, predictive-but-non-causal features

A long-standing intuition in the OOD literature is that models fail under distribution shift because they rely on features that happen to be predictive in the training distribution but are not causally related to the label. [9] consolidate this intuition under the term *shortcut learning*, defining shortcuts as decision rules that perform well on standard benchmarks but fail to transfer to more challenging testing conditions. Their central claim is that shortcut learning is not a property of any single architecture or task but a *general characteristic of how learning systems extract structure from data*, with analogues observable in vision, natural language processing, and reinforcement learning. *The phenomenon is governed by the inductive biases of the learner*, which arise from the interaction of training data, architecture, and optimization procedure, rather than by any single component in isolation.

[10] provide a complementary theoretical account by isolating two mechanisms through which empirical risk minimization comes to rely on spurious features even on tasks where the invariant features are fully predictive of the label. The first is a *geometric skew*: the max-margin classifier preferentially employs spurious features when the invariant features would require a larger-norm separator to achieve the same training margin. The second is a *statistical skew*: gradient descent on logistic loss converges along the spurious direction at a rate proportional to the strength of the spurious correlation, integrating it into the learned solution even when an invariant solution exists. Both mechanisms are demonstrated theoretically for linear classifiers and empirically for neural networks on modified MNIST and CIFAR-10. While there are a variety of ways to tackle statistical skews within ERM using techniques like regularization, data transformation, tree based models etc ... and geometrical skews using power transforms, the shortcomings of statistical skews that could still arise from the input features being skewed and the need of highly specific techniques to tackle these sets of distinct geometric skews motivates us to explore the potential of ICL with demonstration design that is structurally aligned to the true task structure as another technique for optimization.

2.2.2 Shortcut learning across modalities

Although the canonical shortcut-learning examples originate in computer vision (texture bias, background reliance, the Waterbirds and CelebA benchmarks), the phenomenon has been documented across domains. In natural language inference, [11] demonstrate that models trained on MNLI systematically rely on lexical and syntactic heuristics rather than entailment relations, and con-

sequently fail on the HANS adversarial set. In tabular settings, the spurious-correlation problem is closely tied to fairness benchmarks such as Adult, COMPAS, and ACS, where proxy variables (ZIP code, occupation, marital status) remain statistically predictive of the label even after the protected attribute (race, sex) is explicitly excluded from the feature set. A model trained to maximise accuracy will exploit these proxies, producing predictions that are accurate on the training distribution but that encode precisely the demographic shortcuts that fail under population shift. The cross-modality consistency of the phenomenon supports the view that shortcut learning reflects a general principle of feature selection under empirical risk minimization rather than a quirk of any single data type.

A natural question is whether inference-time interventions, particularly demonstration design in in-context learning, can redirect feature reliance without modifying the underlying model. The review returns to this question in Section 2.6.

2.3 ICL and tabular LLMs as the project’s setting

2.3.1 ICL as a learned behaviour

In-context learning, in which a large language model adapts its predictions on the basis of demonstrations placed in its prompt, has been studied from several complementary perspectives [12, 13]. No single account is canonical, and the operative view shapes what kind of intervention on ICL is sensible. Three perspectives are most relevant to the present project.

The **Bayesian inference view**, due to [14], frames ICL as implicit posterior selection over latent concepts. Under this view, the language model treats the prompt as evidence regarding which concept generated the demonstrations and updates its predictive distribution accordingly; in-context learning emerges when the pretraining distribution has a latent-concept structure (formalised as a mixture of hidden Markov models) that supports such inference at test time. The view predicts that demonstration content, not merely demonstration accuracy, governs ICL behaviour, a prediction directly relevant to the project’s interest in demonstration diversity.

The **emergence view** asks when ICL appears at all. [15] demonstrate that ICL is not a generic property of transformers but emerges only when the pretraining distribution exhibits specific structural features: burstiness (items appearing in clusters rather than uniformly), large numbers of rarely occurring classes, and dynamic rather than fixed item meanings. In the absence of these properties, models with identical architectures fail to develop ICL capabilities. [16] extends this empirical finding with a mechanistic account: ICL is implemented by induction heads whose formation during training is abrupt rather than gradual, and the data-distributional dependencies identified by Chan et al. are recapitulated by a minimal two-parameter model of an induction head. The Reddy account traces the abruptness to nested nonlinearities in the attention computation, which create cliffs in the loss landscape. Together, Chan and Reddy

establish that ICL is both data-dependent and dynamically emergent; it is not a capability the model possesses continuously, but one it acquires sharply when the training distribution is appropriate.

The **task-diversity view**, due to [17], addresses a further question: even once ICL has emerged, when does it generalise to tasks outside the pretraining task distribution? Goddard et al. show, in a controlled synthetic regression setting, that out-of-distribution ICL exhibits a phase transition in pretraining task diversity. Below a critical threshold, models produce specialised solutions that succeed only on tasks resembling those seen in pretraining; above the threshold, the same architecture generalises across the full task space. This result constitutes the most direct theoretical motivation for the project's research questions, since it establishes that pretraining task diversity is a sharp lever on OOD ICL, and raises the question, addressed by the present work, of whether an analogous lever exists at inference time through demonstration design.

These three views are not mutually exclusive. They jointly establish that ICL is a learned behaviour whose properties depend on pretraining data structure, the dynamics of induction-head formation, and the diversity of pretraining tasks. Importantly, the theoretical and empirical results above were developed in synthetic or natural-language settings. Whether the same mechanisms govern ICL when a general-purpose LLM receives a tabular prompt, that is, a serialised row of named features and a label, is an open question that the next section addresses.

2.4 ICL on tabular data

2.4.1 Tabular as a distinct regime

The promising properties of in-context learning identified in the previous subsection do not transfer uniformly across data modalities. Two distinct approaches to tabular ICL have emerged, and they differ in the kind of model that processes the demonstrations.

The first approach is *purpose-trained tabular ICL*. [18, 19] introduce TabPFN, a transformer pre-trained exclusively on synthetic tabular tasks generated from a structural-causal-model prior, which performs supervised classification on a new dataset in a single forward pass without per-dataset fitting. TabPFN is positioned as approximating Bayesian inference under its synthetic prior, and on small-data benchmarks it outperforms strong tuned baselines while requiring orders of magnitude less inference time. Crucially, the architecture treats the support set as the in-context demonstrations, which renders TabPFN a clean instantiation of the demonstration-as-context paradigm specialised to the tabular regime. However, TabPFN is not a general-purpose LLM: it possesses no language modelling capability, no pretraining on text, and no ability to process natural-language prompts. Its ICL mechanism is architecturally baked in rather than emergent from language pretraining.

Are we building SATA like TabPFN, if so how is it any different

Are we going to bake in the ICL mech to SATA's architecture but including the previously mentioned metrics, so it identifies the datashifts by measuring the metrics, how would it compare in inference time (when passing the prompt to the frozen LLM) what to compare with ? The metrics are at (*).

SATA is trained on demonstration sets and she picks the best demonstration set to attach with the user query prompt before passing it to frozen LLM in a high level. The demos she trains on are tabular (DETAILS from last sem)

Just training a pre-trained language model / a pre trained transformer would not be enough, be same as TabPFN

The second approach, which is the focus of this project, is *general-purpose LLM ICL applied to tables via prompt serialisation*. Here, a tabular row is converted into a natural-language string (e.g., “Age: 45, Income: \$80k, Education: Bachelor → Approved”), placed in the prompt alongside other serialised rows as demonstrations, and the LLM predicts the query row’s label. This approach inherits whatever ICL capability the LLM acquired during language pretraining, but applies it to a modality, namely structured tabular features with semantic names and mixed types, for which the LLM was not specifically trained. The distinction matters because the failure modes differ: TabPFN’s failures are governed by the coverage of its synthetic prior, whereas general-purpose LLM failures are governed by whatever spurious correlations and distributional properties the language pretraining corpus happened to contain. The project’s demonstration-diversity protocols and meta learned adapter are designed for the latter setting, in which the demonstrations are the primary lever available to the practitioner and the base model is frozen.

2.4.2 Why tabular ICL surfaces OOD failures sharply

The tabular setting brings the OOD failure modes discussed in Section 2.2 into sharp focus for two reasons. First, tabular features possess semantic names and known causal roles in many real domains, which entails that shortcut learning is not a hypothetical concern: a clinically irrelevant but statistically predictive feature (for example, ZIP code as a proxy for race in the ACS benchmark) can be identified, named, and audited, in contrast to the texture features encountered in vision benchmarks. Second, the tabular benchmarks discussed in Section 2.1.3 (TableShift, WHYSHIFT) provide controlled distributions across the shift taxonomy, with the result that the regimes under which ICL succeeds or fails can be characterised precisely. The combination renders general-purpose LLM ICL on tabular data an unusually clean testbed for asking whether demonstration design, as a manipulation external to the base model, can redirect feature reliance under shift. This is the question that the remainder of the review and the project pursue.

2.5 Generalising for the right reasons: from invariance to faithfulness

2.5.1 Invariance and the causal lineage

The intuition that good models should generalise by tracking stable causal mechanisms rather than spurious correlations has a longer history than the ICL literature. [20] formalise this intuition through Invariant Causal Prediction (ICP), which exploits data from multiple environments to identify the predictors whose conditional relationship with the target is invariant across environments. Under their assumptions, this invariant set coincides with the direct causes of the target, providing a statistical procedure with explicit causal semantics. [21] adapt the same intuition to deep learning through Invariant Risk Minimization

(IRM), an objective that searches for representations whose optimal classifier is simultaneously optimal in every training environment. The IRM agenda has become the dominant statement of the “learn from many environments to ignore spurious features” programme.

However, the empirical record of this programme is mixed. [4], discussed earlier, find that IRM and related domain-generalization methods rarely exceed the performance of tuned ERM. [22] and [23] provide formal analyses demonstrating that the practical IRMv1 objective can fail to capture the invariances it is designed to capture, and can generalise worse than ERM on simple problems consistent with the original motivating examples. The conceptual framework is sound; the optimisation objectives currently available to instantiate it on deep models are not reliably effective. This gap matters for the present project because it shows that “generalise for the right reasons” is a difficult problem at the optimisation level, which motivates approaches that target the same goal through different levers, including the inference-time demonstration design the project pursues.

2.5.2 From invariance to faithfulness as evaluation

Invariance and faithfulness are related but not identical, and the distinction matters for what the project can claim. **Invariance concerns whether the learned decision rule remains stable across environments: a model is invariant if its prediction mechanism does not change when the distribution shifts.** Faithfulness, by contrast, concerns whether the model’s stated or measured feature reliance corresponds to the features that actually drive its predictions. **A model can be invariant yet unfaithful (it uses the same wrong features everywhere) or faithful yet non-invariant (it correctly reports shifting to different features in different environments).** The **project’s evaluation targets faithfulness specifically**, because the intervention (demonstration design) operates on a frozen model whose internal decision rule cannot be retrained; what can change is which features the model attends to given different demonstrations.

[24] demonstrate that chain-of-thought reasoning traces can be systematically unfaithful: when biasing features (such as the answer position in a few-shot prompt) are added to the input, models change their predictions to align with the bias and produce CoT explanations that rationalise the biased answer without mentioning the bias itself. [25] extend this with a battery of necessary tests for CoT faithfulness, including sensitivity to perturbations and truncations of the reasoning trace. Both papers operate at the instance level: faithfulness is treated as a property of an individual prediction’s stated reasoning relative to the computation that produced it.

[8] introduce a complementary, domain-level notion. Rather than asking whether a single CoT trace reflects the steps of a single prediction, **STaDS asks whether the model’s self-reported feature ranking across an entire tabular task corresponds to its behavioural feature ranking, where the latter is computed via**

leave-any-out ablation. The Spearman correlation between self-attribution and ablation-based attribution thereby constitutes a measure of *global* faithfulness, distinct from the instance-level CoT faithfulness of Turpin and Lanham. The headline empirical finding, namely that frontier LLMs can be highly accurate yet globally unfaithful, with self-reported reliance diverging from behavioural reliance, is the central evaluation problem the present project inherits. RQ3 in particular is a global-faithfulness question.

2.6 Demonstration design as the lever

2.6.1 Demonstration choice materially affects ICL

A consistent finding across the ICL literature is that the choice of demonstrations is not incidental to performance but a primary driver of it. [26] demonstrate that retrieving demonstrations semantically similar to the query consistently outperforms random selection, with gains of tens of points on table-to-text and open-domain question answering benchmarks. The result establishes demonstrations as a tunable lever: changing only which examples appear in the prompt, while leaving the model and the query untouched, materially shifts predictions.

[27] sharpen the question of what aspect of demonstrations carries this weight. Across twelve language models including GPT-3, the authors find that randomly replacing demonstration labels with incorrect labels barely degrades performance: ground-truth input-label mapping is not what makes ICL work. Instead, three properties matter most: the label space the demonstrations span, the input-text distribution they sample, and the format of input-label pairing. This finding has direct implications for the project. If demonstration labels were the key signal, then demonstration design would reduce to retrieval of correctly-labelled similar examples. Because labels are not the key signal, demonstration design has space to operate on *which regions of input space and which regions of label space the demonstrations cover*, which is precisely what the four diversity protocols (label, feature-range, rule, counter-spurious) are constructed to manipulate.

2.6.2 Gap: demonstration design has not targeted faithful OOD generalisation

The demonstration-selection literature has largely optimized for in-distribution accuracy. The retrieval methods of [26] select demonstrations by semantic similarity to the query and are evaluated primarily on in-distribution benchmarks; while the paper includes a dataset-transfer experiment (SST-2 demonstrations applied to IMDB) and reports gains on the ToTTo nonoverlap subset, distribution shift is not its central concern and is not studied systematically. The analyses of [27] characterize which aspects of demonstrations **drive in-context** learning, **finding that the label space, input distribution, and format** matter far more than the ground-truth input-label mapping, **but conduct this decomp-**

sition on standard benchmarks without varying the test distribution. Neither body of work asks whether demonstration design can be used to improve generalization under shift, and neither evaluates whether the resulting predictions are faithful; that is, whether the model relies on the features it appears to rely on, or whether it achieves accuracy through spurious shortcuts that happen to survive the shift. These gaps, taken together, constitute the gap that the project addresses through its four research questions and a new intervention.

This gap motivates the proposed project: to test whether carefully designed in-context demonstrations can improve both OOD robustness and behavioural faithfulness in tabular LLM prediction. The review has established that OOD failure is driven by shortcut learning (Section 2.2), that ICL is a learned behaviour whose properties depend on pretraining and demonstration structure (Section 2.3), that tabular data provides a uniquely auditable testbed for these questions (Section 2.4), that faithfulness evaluation is necessary because accuracy alone conceals reliance on incorrect features (Section 2.5), and that existing demonstration-selection work has not addressed either OOD robustness or faithfulness (Section 2.6). The project fills this gap by introducing demonstration-diversity protocols targeting specific shift types, a meta-learned adapter that reweights demonstrations in a query-conditioned manner, and an evaluation framework that jointly measures predictive robustness, shift attribution, and decision faithfulness.

3 Task 2: Research Problem and Contributions

Large language models can perform tabular classification through in-context learning, adapting to a new task on the basis of labelled rows placed in the prompt. However, the literature reviewed above exposes three converging gaps. First, shortcut learning causes models to rely on spurious features rather than causally relevant ones, and this failure mode is architecture-agnostic. Second, recent work has shown that in-context learning is itself susceptible to spurious correlations present in the demonstration set [28], and that standard demonstration-selection methods optimise for in-distribution accuracy without considering distributional shift. Third, recent faithfulness work has shown that accuracy alone is an unreliable proxy for understanding: frontier LLMs on tabular tasks can produce correct predictions while relying on the wrong decision factors [8], and standard attribution evaluation methods can themselves be unreliable when perturbations push inputs out of distribution [29]. No existing work examines whether demonstration design, that is, the choice and weighting of in-context examples, can produce out-of-distribution generalisation on tabular tasks that is both accurate and faithful.

The central hypothesis is that out-of-distribution generalisation in tabular in-context learning improves when demonstrations span a richer coverage of decision boundaries, feature interactions, and subpopulation structure; however, predictive gains will not reliably imply faithful factor reliance unless that di-

versity is structurally aligned with the true task mechanism. This hypothesis yields four research questions. **RQ1** asks how well frozen LLMs perform OOD in-context generalisation on structured tabular decision tasks under controlled distribution shifts. **RQ2** asks which kinds of demonstration diversity matter most for OOD performance in this setting. **RQ3** asks whether configurations that improve OOD accuracy also improve global decision faithfulness, or whether accuracy gains can coexist with continued reliance on spurious features. **RQ4** asks whether a small meta-learned adapter operating at the demonstration level can induce a qualitative improvement in both accuracy and faithfulness, analogous to the pretraining-diversity transition identified by [17] but achieved at inference time through demonstration design.

The first contribution is a set of four demonstration-diversity protocols, each targeting a specific failure mode. Recent work has shown that in-context learners are susceptible to spurious features present in the training tasks [28], and that ICL under OOD fails because of spurious correlations carried in context [30], which motivates protocols that explicitly counteract this at the demonstration level. **Label diversity** ensures both classes are represented in the support set, thereby preventing majority-class priors under label shift. **Feature-range diversity** requires demonstrations to span the support of important continuous features, which forces the model to encounter the full input space rather than a narrow cluster that happens to correlate with the label. **Rule diversity** selects demonstrations from distinct decision regimes within the task, for instance a region in which feature A dominates and another in which the interaction $A \times B$ matters, such that no single local shortcut suffices. **Counter-spurious diversity** deliberately includes examples that violate the most common spurious correlations in the data, thereby denying the model the option of relying on them. Each protocol is applied as a controlled experimental condition, compared against random and similarity-based selection under the same distribution shifts.

The second contribution is **SATA** (Shift-Aware Task Adapter), a small trainable transformer that sits alongside the frozen base LLM and learns to reweight in-context demonstrations in a query-conditioned manner. **SATA takes the full support set and the query as input.** Internally, it constructs a task representation from the support set and uses that representation, conditioned on the query, to compute per-demonstration relevance weights. The output from SATA that we want is the score for each demo after the self-attention layers, after these few transformer layers we take each demo position’s output and pass it through a small scoring head, these are then each demonstration’s score. We pick the top k demonstration by assigning a probability score to each via a softmax function. The base LLM’s weights are never modified; the entire intervention operates on what the model receives, rather than on how it processes that input.

SATA’s design draws on the In-Context Risk Minimization framework of [31], which demonstrated that a transformer attending to contextual examples at inference time can learn to extract environment-specific representations that

So are the above libraries we are going to use give us all these as diversified data sets ?

improve OOD generalisation. **SATA adapts this mechanism to the tabular setting:** it is **meta-trained on a synthetic distribution of tabular tasks** spanning linear, threshold, tree-style, and sparse interaction rules, **each instantiated under six shift types** (in-distribution, covariate shift, spurious-correlation reversal, support extrapolation, missing-feature shift, and mechanism shift). The synthetic setting is necessary because ground-truth knowledge of which features should drive predictions is required both to supervise SATA’s reweighting during training and to measure faithfulness during evaluation.

The **evaluation is structured around three axes:** predictive robustness, shift attribution, and decision faithfulness.

Predictive robustness is assessed using the joint robustness-and-uncertainty paradigm of [32]. Rather than reporting accuracy alone, **the evaluation traces error-retention curves:** according to [32] model predictions are progressively replaced by ground-truth labels in order of decreasing uncertainty, and the area under this curve (R-AUC) captures both predictive quality and the informativeness of the model’s uncertainty estimates within a single metric. The evaluation additionally reports F1-retention curves and F1 at 95% retention (F1@95%), which measure the reliability of the model’s most confident predictions. This joint paradigm is necessary because a model that is accurate but poorly calibrated, namely confident on OOD inputs where it is incorrect, would appear strong under accuracy-only evaluation but weak under retention-curve evaluation.

Shift attribution We attribute the accuracy gap to two sources using the procedure of Liu et al. [7]: a covariate component arising from a change in $P(X)$, and a conditional component arising from a change in $P(Y|X)$. We denote these as ΔX and $\Delta Y|X$ respectively. This decomposition is essential for answering RQ2, because different demonstration-diversity protocols target different shift types: feature-range diversity addresses covariate shift, whereas rule diversity and counter-spurious diversity address concept shift. Without shift attribution, a protocol that improves aggregate OOD accuracy could be succeeding on one shift type while failing on another, and the aggregate metric would conceal this asymmetry.

Decision faithfulness measures whether the model relies on the features it claims to rely on. The procedure follows the leave-any-out attribution approach of [8], adapted using the in-distribution perturbation principle of [29]: rather than zeroing or masking an ablated feature, which can push the input out of distribution and distort the model’s behaviour, the procedure replaces each ablated feature’s value with a sample drawn from that feature’s training marginal distribution. For each feature j , the evaluation computes the performance drop Δ_j when feature j is replaced across all test rows, yielding a behavioural importance ranking π_{behav} . The model’s self-reported importance ranking π_{self} is elicited separately by prompting the base LLM to rank features by importance for the prediction task. The Spearman rank correlation $\rho(\pi_{\text{self}}, \pi_{\text{behav}})$ between these two rankings serves as the measure of global decision faithfulness: a high

ρ indicates the model relies on the features it claims to rely on, whereas a low ρ indicates divergence between stated rationale and actual behaviour.

Four baselines isolate the contribution of each component. First, vanilla ICL with random demonstrations establishes whether any demonstration design matters at all. Second, similarity-based retrieval following [26] tests whether the current best-practice selection method suffices for OOD tabular tasks. Third, the four diversity protocols applied without SATA test whether principled demonstration selection alone closes the gap, or whether learned reweighting adds value. Fourth, SATA ablations, namely removing the query-condition so that the scores for each demonstration is consistent for all queries within a task would allow us to identify which aspects of SATA’s design drive the observed gains.

Success is defined per research question to ensure falsifiability. RQ1 succeeds if vanilla ICL shows measurable OOD degradation on a majority of the benchmark settings, as measured by R-AUC and shift gap, thereby establishing that the problem exists in the tabular ICL regime. RQ2 succeeds if at least one diversity protocol significantly outperforms random selection, and if the DISDE decomposition reveals that different protocols improve performance on different shift types, thereby confirming that the distinction between covariate and concept shift matters for demonstration design. RQ3 succeeds if there exist configurations where R-AUC improves but $\rho(\pi_{\text{self}}, \pi_{\text{behav}})$ does not, thereby confirming that predictive gains can coexist with unfaithful feature reliance. RQ4 succeeds if SATA combined with the best-performing protocol outperforms that protocol alone on both R-AUC and $\rho(\pi_{\text{self}}, \pi_{\text{behav}})$, thereby demonstrating that learned reweighting improves both robustness and faithfulness beyond what hand-designed diversity achieves.

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